

NEXUCON INSPECTOR PWA

PART 1: SITE MAP & INFORMATION ARCHITECTURE

1.1 High-Level Architecture

The application is structured as a **Progressive Web App (PWA)** with a **dual-path data ingestion engine** (Telemetry + Manual Import) that syncs to the Government Dashboard.

```
graph TD
  A[Nexucon Inspector PWA] --> B[Core Modules]
  A --> C[Data Ingestion Engine]
  A --> D[Sync Layer]

  B --> B1[Command Center]
  B --> B2[Assigned Projects]
  B --> B3[Field Inspections]
  B --> B4[Digital Eye]
  B --> B5[Evidence Vault]
  B --> B6[Findings & SWOs]

  C --> C1[Telemetry Path]
  C --> C2[Manual Import Path]
```

C1 --> C1a[GPR Radargram]

C1 --> C1b[UPV Device]

C1 --> C1c[Tersus SLAM]

C1 --> C1d[Thermal Camera]

C2 --> C2a[CSV Upload]

C2 --> C2b[PDF Upload]

C2 --> C2c[JSON Upload]

D --> D1[IndexedDB Local Cache]

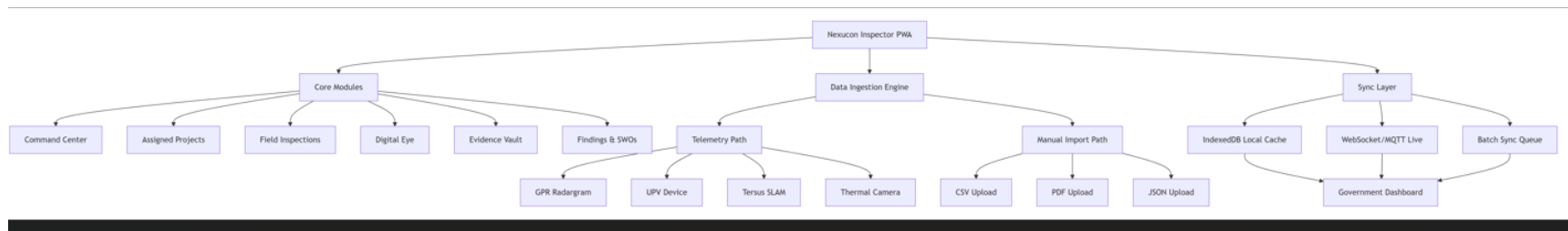
D --> D2[WebSocket/MQTT Live]

D --> D3[Batch Sync Queue]

D1 --> E[Government Dashboard]

D2 --> E

D3 --> E



1.2 Detailed Site Map

Level 1	Level 2	Level 3	Route	Purpose
Oversight & Field	Command Center		/inspector/dashboard	Cockpit view
	Assigned Projects		/inspector/dashboard/projects	List of assigned sites
		Project Detail	/inspector/dashboard/projects/[id]	Project overview
	Field Inspections		/inspector/dashboard/inspections	List of inspections
		Inspection Terminal	/inspector/dashboard/inspections/[id]	Active inspection
		Checklist Stage	/inspector/dashboard/inspections/[id]/stage/[stageId]	Stage-specific checklist

Level 1	Level 2	Level 3	Route	Purpose
Technical Analysis	Evidence Vault		/inspector/dashboard/evidence	Immutable media ledger
	Digital Eye		/inspector/dashboard/digital-eye	Hub for NDT/GPR/BIM
		GPR Radargram Viewer	/inspector/dashboard/digital-eye/gpr	Subsurface scan analysis
		UPV Analyzer	/inspector/dashboard/digital-eye/upv	Pulse velocity analysis
		Tersus SLAM Viewer	/inspector/dashboard/digital-eye/slam	3D point cloud

Level 1	Level 2	Level 3	Route	Purpose
		BIM Deviation Viewer	/inspector/dashboard/digital-eye/bim	As-built vs design
	Findings & SWOs		/inspector/dashboard/findings	Defect lifecycle
		Finding Detail	/inspector/dashboard/findings/[id]	Single finding view
		Issue SWO	/inspector/dashboard/findings/[id]/swo	Stop Work Order
	Compliance Standards		/inspector/dashboard/compliance	Regulatory reference

**Docs &
System**

Level 1	Level 2	Level 3	Route	Purpose
	Site Documents		/inspector/dashboard/documents	Project files
	Inspection Reports		/inspector/dashboard/reports	Generated reports
	Alerts & Notices		/inspector/dashboard/notifications	System alerts
	Inspector Profile		/inspector/dashboard/settings	User settings
Sync & Data				
	Sync Center		/inspector/dashboard/sync	Sync status & queue
	Manual Import		/inspector/dashboard/sync/import	CSV/JSON/PDF upload

Level 1	Level 2	Level 3	Route	Purpose
	Telemetry Status		/inspector/dashboard/sync/telemetry	Live device status

1.3 Navigation Structure

Desktop Sidebar (Collapsible):

text

NEXUCON INSPECTOR
 Command Center  Assigned Projects  Field Inspections  Evidence Vault
TECHNICAL ANALYSIS
 Digital Eye (GPR/BIM)  Findings & SWOs  Compliance Standards
DOCS & SYSTEM
 Site Documents

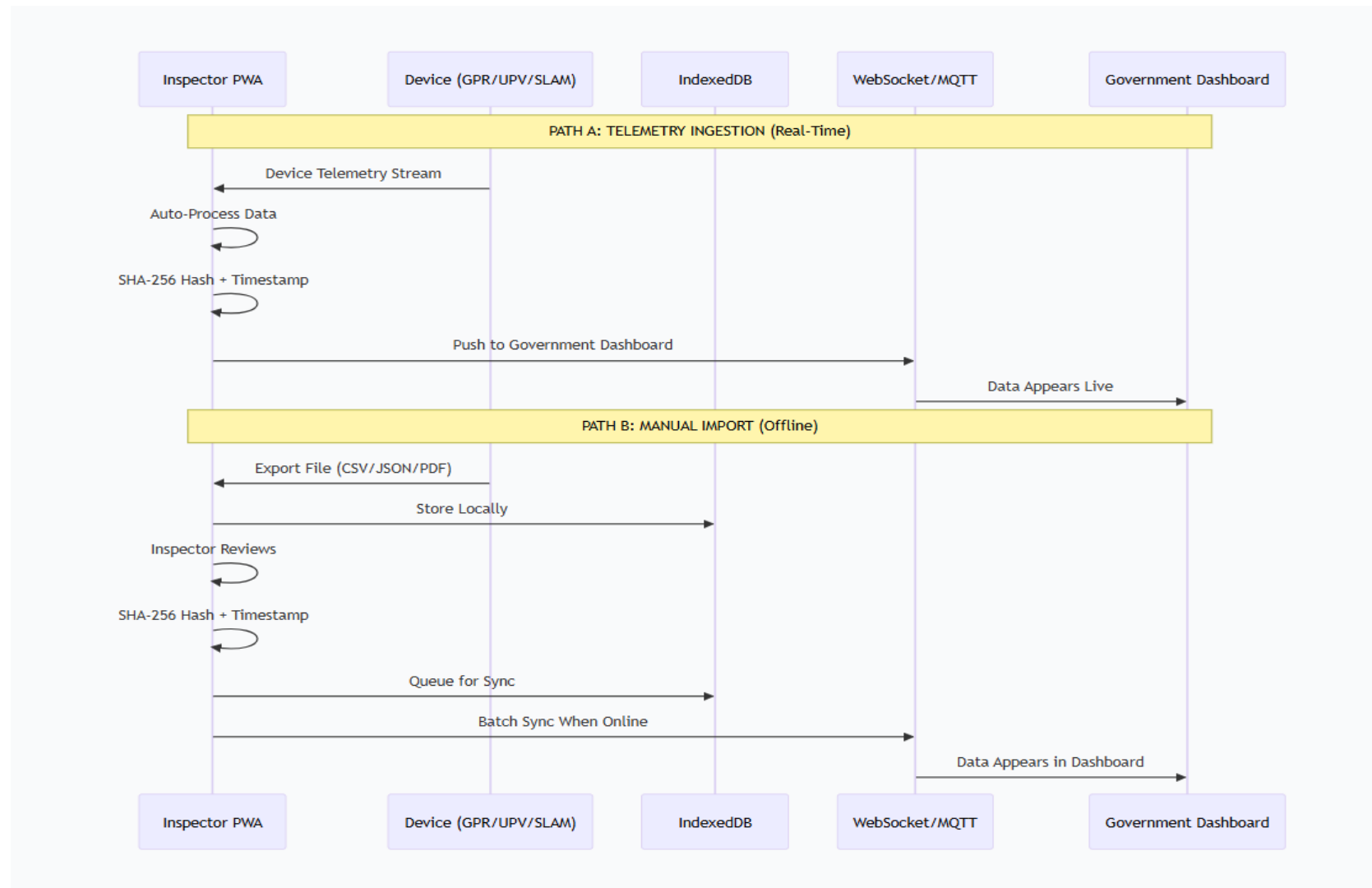
		 Inspection Reports		
		 Alerts & Notices		
		 Inspector Profile		
		SYNC CENTER		
		 Sync Status		
		 Manual Import		
	[Live Sync Active 		[Offline Cache]	

Mobile Bottom Navigation:

text

	[🏠 Home]	[📁 Projects]	[📂 Inspect]	[📷 Evidence]	[≡]

1.4 Data Flow: Dual-Path Ingestion



PART 2: WIREFRAMES FOR THE 6 CORE MODULES

Module 1: Daily Scheduling & Dispatch

Route: `/inspector/dashboard` (Command Center) + `/inspector/dashboard/projects`

text

COMMAND CENTER

Inspector: Badge #LAG-INS-042 • Ikeja North Directorate

[Live Sync Active ] [3 Pending Sync 

Assigned Sites	Upcoming Tasks	Open Findings	Non-Compliances	Pending Sync
12	5	8	3	3

TODAY'S FIELD INSPECTION SCHEDULE

 09:00	Eko Atlantic Tower	Lekki Phase 1	[CHECK IN]
 11:30	Landmark Towers	Victoria Island	[CHECK IN]
 14:00	Chief Magistrate	Ebute Metta	[CHECK IN]

DIGITAL EYE RADAR BANNER

 GPR SCAN
Subsurface radar

 UPV ANALYZER
Pulse velocity

 BIM DEVIATION
3D laser scan

CRITICAL FINDINGS & STOP WORK ORDERS

 STOP-WORK ORDER: Eko Atlantic Tower - Sector 4
Reason: Structural deviation >10mm on Column C4
Issued: 2 days ago | Status: ACTIVE | [\[VIEW DETAILS\]](#)

EVIDENCE INTEGRITY WIDGET

✔ 247 files SHA-256 verified | 📁 3 files pending sync

Module 2: Geofenced GPS Site Arrival

Route: `/inspector/dashboard/inspections/[id]` (Check-In)

text

FIELD INSPECTION - EKO ATLANTIC TOWER

← Back

GEOFENCED GPS CHECK-IN

[INTERACTIVE MAP]

● Your Location (6.4281° N, 3.4219° E)

 Site Boundary (50m radius)

Distance: 32m | Status: WITHIN GEOFENCE ✓

SITE ADDRESS: Eko Atlantic Tower, Lekki Phase 1, Lagos

GPS COORDINATES: 6.428100° N, 3.421900° E

INSPECTOR: Badge #LAG-INS-042		
TIMESTAMP: 2026-09-16 09:15:23 WAT		
[📷 TAKE SITE PHOTO] [📍 CAPTURE GPS + CHECK IN]		
⚠️ WARNING: You must be within 50m of the site to check in.		
Current distance: 32m ✓		

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Module 3: Interactive Multi-Stage Structural Checklists

Route: /inspector/dashboard/inspections/[id]/stage/[stageId]

text

STAGE 1: FOUNDATION EXCAVATION	[← Back] [Save Draft]
PROGRESS: 3/5 items completed	[██████████] 60%

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CHECKLIST ITEMS

1. Foundation Excavation Depth & Bearing Strata Check

[✓ PASS] [✗ FAIL] [— N/A]

Comments: [Excavation depth is 2.5m, matches design]

[📷 Add Photo] [🎤 Voice Note]

2. Rebar Spacing, Diameter & Splice Length Compliance

[✓ PASS] [✗ FAIL] [— N/A]

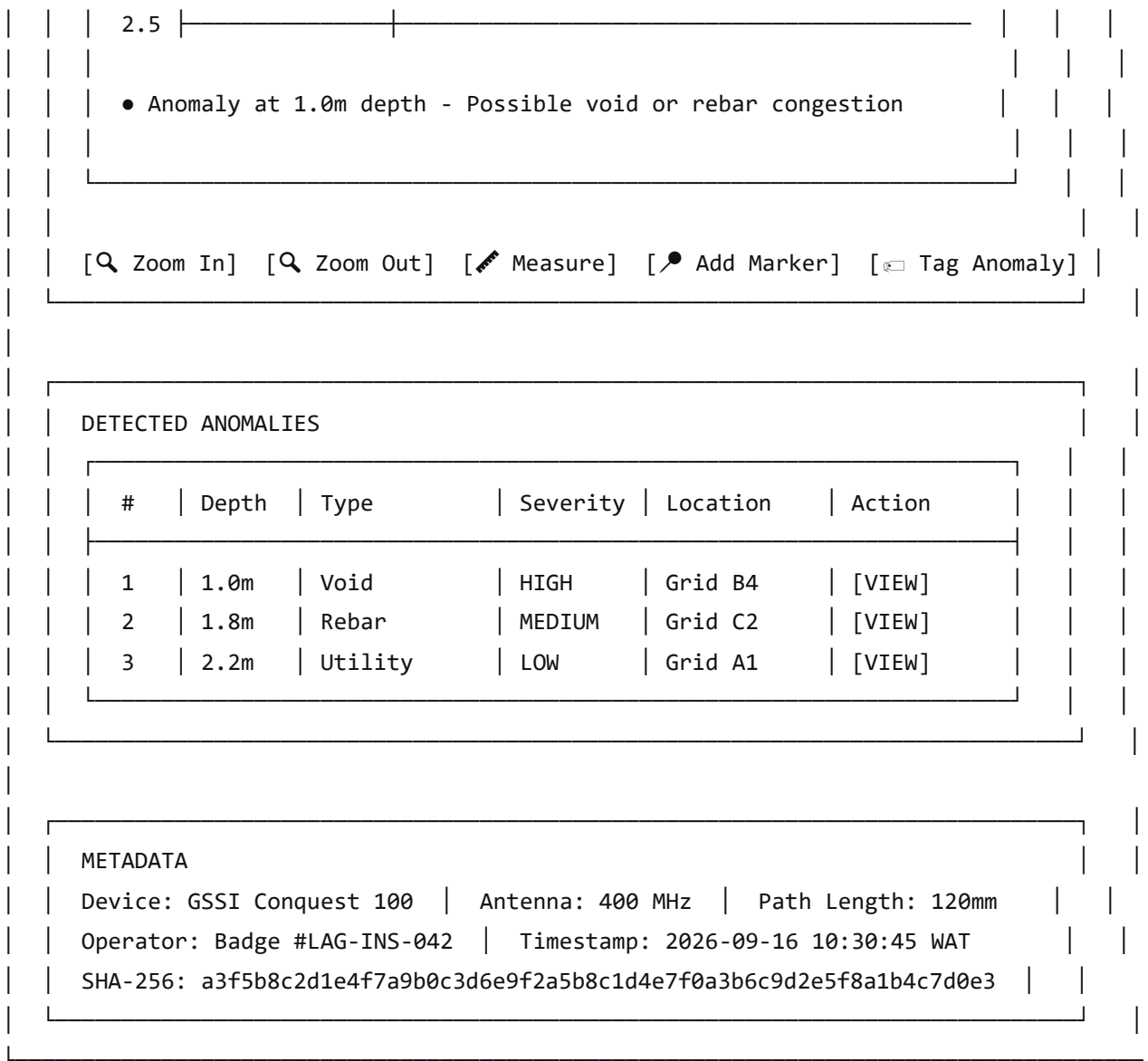
Comments: [Spacing is 150mm, within tolerance]

[📷 Add Photo] [🎤 Voice Note]

3. Concrete Cover Spacers & Formwork Alignment

[✓ PASS] [✗ FAIL] [— N/A]

		Comments: [Cover spacers in place, formwork aligned]			
		[📷 Add Photo] [🗣️ Voice Note]			
		4. Damp-Proof Membrane & Subsurface Drainage Verification			
		[✅ PASS] [❌ FAIL] [— N/A]			
		Comments: [Not yet installed - pending next stage]			
		[📷 Add Photo] [🗣️ Voice Note]			
		5. Site Safety Scaffolding, Edge Protection & PPE Review			
		[✅ PASS] [❌ FAIL] [— N/A]			
		Comments: [Safety compliance verified]			
		[📷 Add Photo] [🗣️ Voice Note]			
		[⚠️ LOG DEFECT] [👤 ADD COMMENT] [✅ SUBMIT STAGE]			



Module 5: NDT Data Capture (UPV + Tersus SLAM)

Route: /inspector/dashboard/digital-eye/upv

text

UPV ANALYZER (ULTRASONIC PULSE VELOCITY)

[\[← Back\]](#)[\[Export\]](#)

DATA SOURCE:

 Live Telemetry

 Manual Import

LIVE MEASUREMENT

Path Length (mm):

[120]

Transit Time (μs):

[30.1]

Temperature (°C):

[28]

Rebound Number:

[32]

 (for SonReb)

 CAPTURE FROM DEVICE

 MANUAL ENTRY

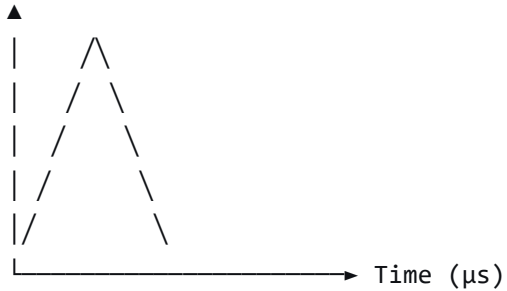
LOCKED OUTPUT ENGINE (Server-Computed)

Pulse Velocity (V): 4,000 m/s
Estimated Strength: 27.9 MPa
Threshold (25 MPa): ✓ PASSED
Confidence: 95% (CI: 26.3 - 29.5 MPa)
Formula: $f_{cu} = 8.961 \times V - 7.97$ (E.C.S calibration)

WAVEFORM OSCILLOGRAM

[WAVEFORM DISPLAY]

Amplitude



[▶ PLAY] [⏸ PAUSE] [📏 MEASURE] [🔍 ZOOM]

TEST POINTS REGISTRY									
Point	Path	Time	Velocity	Strength	Status	Timestamp			
A	120	30.1	3,986	27.8	GOOD	10:30:45			
B	120	29.8	4,026	28.1	GOOD	10:32:15			
C	120	30.3	3,960	27.5	GOOD	10:34:00			
[📄 SAVE MEASUREMENT] [📤 PUSH TO DASHBOARD] [📂 EXPORT]									

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Module 6: Cryptographic Evidence Validation

Route: /inspector/dashboard/evidence

text

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EVIDENCE VAULT

← Back

Upload

INTEGRITY STATUS:

✓ 247 files SHA-256 verified

 |

🔄 3 files pending sync

FILTERS:

All ▼

Photos

Thermal

NDT Files

Documents

MEDIA LEDGER

📷	Column C4 Crack.jpg	SHA-256: a3f5...	✓ VERIFIED
	GPS: 6.4281, 3.4219	2026-09-16 10:30	
🌡️	Thermal_Scan_01.jpg	SHA-256: b8c2...	✓ VERIFIED
	GPS: 6.4281, 3.4219	2026-09-16 10:35	
📊	UPV_Reading_01.csv	SHA-256: c1d4...	✓ VERIFIED
	GPS: 6.4281, 3.4219	2026-09-16 10:40	
📡	GPR_Radargram_01.dzt	SHA-256: d4e7...	🔄 PENDING
	GPS: 6.4281, 3.4219	2026-09-16 11:00	

SELECTED FILE DETAILS		
File: Column C4 Crack.jpg		
Type: Photo		
Size: 2.4 MB		
GPS: 6.428100° N, 3.421900° E		
Timestamp: 2026-09-16 10:30:45 WAT		
Inspector: Badge #LAG-INS-042		
SHA-256: a3f5b8c2d1e4f7a9b0c3d6e9f2a5b8c1d4e7f0a3b6c9d2e5f8a1b4c7d0e3		
Status:  VERIFIED (Tamper-proof)		
[ VERIFY HASH] [ SYNC TO DASHBOARD] [ EXPORT] [ DELETE]		

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PART 3: DATA SCHEMA FOR TELEMETRY + MANUAL IMPORT

3.1 Unified Data Schema (Both Paths Land Here)

sql

```
-- =====
-- CORE INSPECTION TABLES
-- =====

-- Inspectors Table
CREATE TABLE inspectors (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  badge_number VARCHAR(50) UNIQUE NOT NULL,
  full_name VARCHAR(200) NOT NULL,
  directorate VARCHAR(100) NOT NULL,
  email VARCHAR(200) UNIQUE NOT NULL,
  phone VARCHAR(50),
  accreditation_status VARCHAR(20) DEFAULT 'ACTIVE', -- ACTIVE, SUSPENDED, EXPIRED
  accreditation_expiry DATE,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- Projects Table (Linked to Government Dashboard)
CREATE TABLE projects (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  project_code VARCHAR(50) UNIQUE NOT NULL,
  project_name VARCHAR(200) NOT NULL,
  developer_name VARCHAR(200),
```

```

contractor_name VARCHAR(200),
site_address TEXT NOT NULL,
site_lat FLOAT NOT NULL,
site_long FLOAT NOT NULL,
geofence_radius_m INTEGER DEFAULT 50,
project_status VARCHAR(50) DEFAULT 'ACTIVE',
created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

-- Inspections Table

```

CREATE TABLE inspections (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  project_id UUID NOT NULL REFERENCES projects(id),
  inspector_id UUID NOT NULL REFERENCES inspectors(id),
  inspection_type VARCHAR(50) NOT NULL, -- FOUNDATION, COLUMNS, SLAB, etc.
  stage VARCHAR(50), -- Stage 1, Stage 2, etc.
  scheduled_date TIMESTAMP WITH TIME ZONE,
  check_in_time TIMESTAMP WITH TIME ZONE,
  check_in_lat FLOAT,
  check_in_long FLOAT,
  check_out_time TIMESTAMP WITH TIME ZONE,
  status VARCHAR(50) DEFAULT 'SCHEDULED', -- SCHEDULED, IN_PROGRESS, COMPLETED, FAILED
  outcome VARCHAR(50), -- PASSED, CONDITIONAL_PASS, FAILED
  notes TEXT,
  sha256_hash VARCHAR(64), -- Cryptographic hash of the inspection record
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

```
);
```

```
-- Checklist Items Table
```

```
CREATE TABLE checklist_items (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  inspection_id UUID NOT NULL REFERENCES inspections(id),  
  item_order INTEGER NOT NULL,  
  item_description TEXT NOT NULL,  
  result VARCHAR(20), -- PASS, FAIL, NA  
  comments TEXT,  
  photo_url VARCHAR(500),  
  voice_note_url VARCHAR(500),  
  sha256_hash VARCHAR(64),  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP  
);
```

```
-- Findings Table
```

```
CREATE TABLE findings (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  inspection_id UUID NOT NULL REFERENCES inspections(id),  
  finding_code VARCHAR(50) UNIQUE NOT NULL,  
  category VARCHAR(50) NOT NULL, -- STRUCTURAL, SAFETY, QUALITY, PERMIT  
  severity VARCHAR(20) NOT NULL, -- LOW, MEDIUM, HIGH, CRITICAL  
  location_description TEXT,  
  location_lat FLOAT,  
  location_long FLOAT,  
  description TEXT NOT NULL,  
  recommended_action TEXT,
```

```

    status VARCHAR(50) DEFAULT 'OPEN', -- OPEN, UNDER_REVIEW, CORRECTIVE_ACTION_PENDING, RESOLVED
    assigned_to UUID,
    due_date DATE,
    sha256_hash VARCHAR(64),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

-- Stop Work Orders Table

```

CREATE TABLE stop_work_orders (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    swo_code VARCHAR(50) UNIQUE NOT NULL,
    finding_id UUID NOT NULL REFERENCES findings(id),
    project_id UUID NOT NULL REFERENCES projects(id),
    issued_by UUID NOT NULL REFERENCES inspectors(id),
    issued_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    reason TEXT NOT NULL,
    regulatory_citation TEXT,
    status VARCHAR(50) DEFAULT 'ACTIVE', -- ACTIVE, LIFTED, DISPUTED
    lifted_at TIMESTAMP WITH TIME ZONE,
    lifted_by UUID REFERENCES inspectors(id),
    lift_reason TEXT,
    sha256_hash VARCHAR(64),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

3.2 Telemetry Ingestion Schema

sql

```
-- =====  
-- TELEMETRY INGESTION TABLES  
-- =====  
  
-- Device Registry  
CREATE TABLE devices (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
    device_type VARCHAR(50) NOT NULL, -- GPR, UPV, SLAM, THERMAL  
    device_serial VARCHAR(100) UNIQUE NOT NULL,  
    manufacturer VARCHAR(100),  
    model VARCHAR(100),  
    calibration_date DATE,  
    calibration_expiry DATE,  
    status VARCHAR(20) DEFAULT 'ACTIVE',  
    last_telemetry TIMESTAMP WITH TIME ZONE,  
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP  
);  
  
-- Telemetry Sessions  
CREATE TABLE telemetry_sessions (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
    device_id UUID NOT NULL REFERENCES devices(id),  
    inspection_id UUID NOT NULL REFERENCES inspections(id),  
    session_start TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
```

```

    session_end TIMESTAMP WITH TIME ZONE,
    data_payload JSONB NOT NULL, -- Raw telemetry data
    sha256_hash VARCHAR(64),
    sync_status VARCHAR(20) DEFAULT 'PENDING', -- PENDING, SYNCED, FAILED
    synced_at TIMESTAMP WITH TIME ZONE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

-- GPR Data

```

CREATE TABLE gpr_data (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    telemetry_session_id UUID REFERENCES telemetry_sessions(id),
    inspection_id UUID NOT NULL REFERENCES inspections(id),
    file_url VARCHAR(500) NOT NULL,
    file_format VARCHAR(20), -- .dzt, .rd3, .segy
    antenna_frequency_mhz INTEGER,
    path_length_mm INTEGER,
    scan_depth_m FLOAT,
    anomalies JSONB, -- Detected anomalies
    sha256_hash VARCHAR(64),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

-- UPV Data

```

CREATE TABLE upv_data (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    telemetry_session_id UUID REFERENCES telemetry_sessions(id),
    inspection_id UUID NOT NULL REFERENCES inspections(id),

```

```

    point_id VARCHAR(50) NOT NULL,
    path_length_mm FLOAT NOT NULL,
    transit_time_us FLOAT NOT NULL,
    pulse_velocity_m_s FLOAT NOT NULL,
    temperature_c FLOAT,
    rebound_number FLOAT,
    estimated_strength_n_mm2 FLOAT,
    curve_id UUID, -- Reference to the conversion curve used
    formula_snapshot JSONB,
    confidence_interval_lower FLOAT,
    confidence_interval_upper FLOAT,
    sha256_hash VARCHAR(64),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

-- *SLAM/Tersus Data*

```

CREATE TABLE slam_data (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    telemetry_session_id UUID REFERENCES telemetry_sessions(id),
    inspection_id UUID NOT NULL REFERENCES inspections(id),
    file_url VARCHAR(500) NOT NULL, -- .las, .ply, .pcd
    file_format VARCHAR(20),
    point_count BIGINT,
    bounding_box JSONB,
    deviation_from_bim_mm FLOAT,
    sha256_hash VARCHAR(64),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

```
-- Thermal Data
CREATE TABLE thermal_data (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  telemetry_session_id UUID REFERENCES telemetry_sessions(id),
  inspection_id UUID NOT NULL REFERENCES inspections(id),
  file_url VARCHAR(500) NOT NULL,
  file_format VARCHAR(20), -- .jpg, .png
  anomaly_type VARCHAR(50), -- DELAMINATION, MOISTURE, INSULATION
  temperature_variance_c FLOAT,
  severity VARCHAR(20), -- LOW, MEDIUM, HIGH
  location_lat FLOAT,
  location_long FLOAT,
  sha256_hash VARCHAR(64),
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

3.3 Manual Import Schema

sql

```
-- =====
-- MANUAL IMPORT TABLES
-- =====

-- Import Batches
CREATE TABLE import_batches (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```

    inspector_id UUID NOT NULL REFERENCES inspectors(id),
    inspection_id UUID REFERENCES inspections(id),
    import_type VARCHAR(20) NOT NULL, -- CSV, JSON, PDF
    file_name VARCHAR(200) NOT NULL,
    file_size_bytes BIGINT,
    file_url VARCHAR(500) NOT NULL,
    import_status VARCHAR(20) DEFAULT 'PENDING', -- PENDING, VALIDATED, IMPORTED, FAILED
    validation_errors JSONB,
    imported_at TIMESTAMP WITH TIME ZONE,
    sha256_hash VARCHAR(64),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

-- Import Records (Individual rows from CSV/JSON)

```

CREATE TABLE import_records (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    batch_id UUID NOT NULL REFERENCES import_batches(id),
    record_type VARCHAR(50) NOT NULL, -- UPV, GPR, SLAM, FINDING
    record_data JSONB NOT NULL,
    validation_status VARCHAR(20) DEFAULT 'PENDING',
    error_message TEXT,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

-- Sync Queue

```

CREATE TABLE sync_queue (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    inspector_id UUID NOT NULL REFERENCES inspectors(id),

```

```

entity_type VARCHAR(50) NOT NULL, -- INSPECTION, FINDING, EVIDENCE, TELEMETRY
entity_id UUID NOT NULL,
action VARCHAR(20) NOT NULL, -- CREATE, UPDATE, DELETE
payload JSONB NOT NULL,
sha256_hash VARCHAR(64),
sync_status VARCHAR(20) DEFAULT 'PENDING', -- PENDING, SYNCING, SYNCED, FAILED
retry_count INTEGER DEFAULT 0,
last_error TEXT,
created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
synced_at TIMESTAMP WITH TIME ZONE
);

```

```
-- Evidence Vault
```

```

CREATE TABLE evidence (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  inspection_id UUID NOT NULL REFERENCES inspections(id),
  finding_id UUID REFERENCES findings(id),
  evidence_type VARCHAR(50) NOT NULL, -- PHOTO, THERMAL, NDT_FILE, DOCUMENT
  file_name VARCHAR(200) NOT NULL,
  file_url VARCHAR(500) NOT NULL,
  file_size_bytes BIGINT,
  mime_type VARCHAR(100),
  gps_lat FLOAT,
  gps_long FLOAT,
  captured_at TIMESTAMP WITH TIME ZONE NOT NULL,
  inspector_id UUID NOT NULL REFERENCES inspectors(id),
  sha256_hash VARCHAR(64) NOT NULL,
  verification_status VARCHAR(20) DEFAULT 'PENDING', -- PENDING, VERIFIED, FAILED
);

```

```

    verified_at TIMESTAMP WITH TIME ZONE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- Audit Trail (Immutable)
CREATE TABLE audit_trail (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    entity_type VARCHAR(50) NOT NULL,
    entity_id UUID NOT NULL,
    action VARCHAR(50) NOT NULL,
    old_data JSONB,
    new_data JSONB,
    performed_by UUID NOT NULL REFERENCES inspectors(id),
    ip_address VARCHAR(50),
    user_agent TEXT,
    sha256_hash VARCHAR(64) NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

3.4 Conversion Curves Schema (For UPV)

```

sql

-- =====
-- CONVERSION CURVES (Nexucon Link)
-- =====

CREATE TABLE conversion_curves (

```

```

id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
name VARCHAR(200) NOT NULL,
curve_type VARCHAR(20) NOT NULL, -- LINEAR, POLYNOMIAL, EXPONENTIAL, SONREB, LOOKUP
standard VARCHAR(100),
formula_params JSONB NOT NULL,
-- Examples:
-- Linear: {"m": 0.018, "c": -40}
-- polynomial: {"coeffs": [35, -0.02, 0.000005]}
-- exponential: {"a": 0.0012, "b": 0.0018, "c": 0}
-- sonreb: {"a": 0.00001, "b": 1.2, "c": 1.5}
-- Lookup: {"points": [{"v": 2000, "f": 10}, ...]}
valid_range_min FLOAT,
valid_range_max FLOAT,
is_default BOOLEAN DEFAULT FALSE,
is_active BOOLEAN DEFAULT FALSE,
created_by UUID REFERENCES inspectors(id),
created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

3.5 API Contract: Telemetry & Manual Import

Telemetry Ingestion Endpoints:

Method	Endpoint	Purpose	Payload
POST	/api/v1/telemetry/session/start	Start a telemetry session	{ device_id, inspection_id }
POST	/api/v1/telemetry/session/{id}/data	Push telemetry data	{ data_type, payload }
POST	/api/v1/telemetry/session/{id}/end	End session and hash	{ session_id }
GET	/api/v1/telemetry/session/{id}/status	Get sync status	-
WebSocket	/ws/telemetry/{device_id}	Real-time stream	Binary/JSON

Manual Import Endpoints:

Method	Endpoint	Purpose	Payload
POST	/api/v1/import/upload	Upload CSV/JSON/PDF	multipart/form-data
POST	/api/v1/import/{batch_id}/validate	Validate import	-
POST	/api/v1/import/{batch_id}/commit	Commit to database	-

Method	Endpoint	Purpose	Payload
GET	/api/v1/import/{batch_id}/status	Get import status	-
GET	/api/v1/import/templates/{type}	Get CSV template	-

Evidence Endpoints:

Method	Endpoint	Purpose	Payload
POST	/api/v1/evidence/upload	Upload evidence	multipart/form-data
POST	/api/v1/evidence/{id}/verify	Verify SHA-256 hash	-
GET	/api/v1/evidence/{id}	Get evidence details	-
GET	/api/v1/evidence/inspection/{inspection_id}	List evidence for inspection	-

Sync Endpoints:

Method	Endpoint	Purpose	Payload
POST	/api/v1/sync/queue	Add to sync queue	{ entity_type, entity_id, action, payload }
POST	/api/v1/sync/process	Process queue	-
GET	/api/v1/sync/status	Get sync status	-

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PART 4: SUMMARY OF DELIVERABLES

#	Deliverable	Status	Description
1	Site Map / Information Architecture	✔ Complete	Full navigation structure with 5 levels

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#	Deliverable	Status	Description
2	Wireframes for 6 Core Modules	✓ Complete	Scheduling, GPS Check-In, Checklists, GPR, UPV, Evidence
3	Data Schema for Telemetry + Manual Import	Complete	20+ tables covering all data types
4	API Contract	Complete	15+ endpoints for telemetry, import, evidence, sync

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Part 2: Production-Ready Frontend Code with the following deliverables:

1. **React + TypeScript Component Library** – All 6 modules
2. **PWA Configuration** – Service worker, manifest, offline caching
3. **IndexedDB Schema** – Local storage for offline-first

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4. **API Integration Layer** – Telemetry + Manual Import
5. **SHA-256 Hashing Utilities** – For evidence validation
6. **High-Contrast Theme** – Field-tested UI
7. **Bottom Navigation** – Mobile-first navigation
8. **Data Export** – CSV/JSON for Government Dashboard

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